



tiltdens: Tilted and Data-Sharpened Density Estimation in R

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Abstract

A kernel density estimator built from a non-negative kernel cannot converge faster than $O_p(n^{-4/5})$, however smooth the underlying density may be. Infinite-order kernels break that barrier, but the estimates they produce take negative values and oscillate in the tails: they are not densities, and they suggest structure that is not present. **tiltdens** implements a resolution of this tension, in which a conventional kernel estimator is perturbed toward an infinite-order one, either by re-weighting the observations or by moving them. The result inherits the faster rate while remaining a proper, non-negative density.

The package implements the two criteria of [Doosti and Hall \(2016\)](#) and [Doosti, Hall, and Mateu \(2018\)](#). We show that both reduce to the same convex quadratic program over the probability simplex, differing only in a single linear term, so that one solver serves both and the solution is unique. That reformulation is what makes the methods practical, and it extends without modification to blocked weights, shape constraints and multivariate data. No implementation of tilted density estimation was previously available in any language.

Keywords: tilting, data sharpening, density estimation, quadratic programming, infinite-order kernels, R.

1. Introduction

Let $X_1, \dots, X_n$ be a random sample from a density f , and let

$$\tilde{f}(x | h) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right) \quad (1)$$

be the conventional kernel estimator. When K is a bounded, symmetric probability density and f has two bounded derivatives, $\tilde{f}$ converges at rate $O_p(n^{-4/5})$ in integrated squared error,

and this cannot be improved however smooth f is (Stone 1980). Faster rates require a kernel of higher order, which necessarily takes negative values on part of its support.

The price is well known. A higher-order estimate is not a density: it dips below zero, most visibly in the tails, and it oscillates there in a way that invites the reader to see modes that are not present. Practitioners consequently tend to prefer conventional kernels even when the rate penalty is understood, and Doosti and Hall (2016) open by observing exactly this.

1.1. Perturbed estimators

The resolution proposed by Doosti and Hall (2016) is to keep the conventional kernel and perturb the estimator toward a higher-order one. Two perturbations are available. *Tilting* re-weights the observations,

$$\hat{f}(x \mid h, p) = \sum_{i=1}^n p_i K_h(x - X_i), \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad (2)$$

where $K_h(u) = h^{-1}K(u/h)$, and *data sharpening* moves them,

$$\hat{f}(x \mid h, q) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i - q_i}{h}\right). \quad (3)$$

Because K is a proper density and p is a probability vector, the estimator at (2) is non-negative and integrates to one by construction; the same holds for (3). The perturbation is chosen so that the result sits as close as possible to a comparator $\check{f}$ built from an infinite-order kernel, which transfers the comparator’s convergence rate without its defects. Tilting in this sense goes back to Grenander (1956), and was developed for constrained function estimation by Hall and Presnell (1999) and Hall and Huang (2002).

The 2016 construction is accurate but expensive, since it requires the comparator to be computed and the distance to it minimised. Doosti *et al.* (2018) remove the comparator entirely, choosing h and p together by minimising a cross-validation criterion. That is the method we expect most users to want, and it is the package’s recommended default.

1.2. Existing software

Kernel density estimation is well served in R. **ks** (Duong 2007) and **KernSmooth** provide conventional and higher-order estimators with a range of bandwidth selectors; **sm** and **kedd** cover similar ground. None of them offers any perturbation machinery, so none can deliver a higher-order rate with a guaranteed non-negative estimate.

Shape-constrained estimation is also available, but by a different route. **logcondens** (Dümbgen and Rufibach 2011) maximises a likelihood under log-concavity, which imposes shape by restricting the estimator class rather than by re-weighting observations.

The package closest in name is **sharpData**, which implements data sharpening following Choi, Hall, and Rousson (2000). That work concerns *local linear regression*, not density estimation, and the two problems share only the sharpening idea.

To our knowledge no package, in R or in any other language, implements tilted density estimation.

1.3. Contribution

We separate two things. The first is an implementation of two published methods that previously had none. The second, and the reason a software paper is warranted rather than a code release, is what had to be added to make those methods usable. In order of importance:

1. *One convex program.* Both selection criteria reduce to the same quadratic program over the probability simplex, differing only in one linear term, and the quadratic form is positive semidefinite for any kernel (Section 3). The reformulation gives a unique solution from a standard solver, in place of a search over a high-dimensional simplex.
2. *Optimised block boundaries* (Section 3.1). The 2016 paper treats the boundaries between blocks of equal weight as part of the optimisation but does not say how; reading block sums off a two-dimensional cumulative sum makes exhaustive search affordable. A second placement rule, at the troughs of a pilot estimate, follows a suggestion in the 2018 paper.
3. *Shape constraints* (Section 3.2), following Hall and Huang (2002), and their combination with the cross-validation criterion, which the 2018 paper suggests but does not develop.
4. *Multivariate data* (Section 3.3). Described in Section 3.3 of the 2016 paper; no code existed.
5. *A choice of kernel, with comparable bandwidths* (Section 3.4). The papers use the Gaussian throughout. The package offers the Laplace-convolution family for which the 2016 theory is stated, and the usual compact kernels, with bandwidths rescaled by the canonical factor of Marron and Nolan (1988) so that the choice of kernel changes the shape of the estimate rather than its degree of smoothing.
6. *Numerical details that the papers leave implicit* (Section 3.5): quadrature for the oscillatory cross terms that adapts to the frequency, and a bounded search for the comparator bandwidth that avoids the spurious minima of an infinite-order cross-validation criterion.

Sections 4 and 5 demonstrate the package, and Section 6 places it among existing alternatives.

2. Perturbed density estimation

2.1. Comparator estimators

Both criteria refer to an estimator built from a kernel of unlimited order. The package provides two. The *sinc* estimator uses $L(u) = \sin(u)/(\pi u)$, whose Fourier transform is the indicator of $[-1, 1]$; the *trapezoidal* or flat-top estimator of Politis (2003) and Politis and Romano (1999) uses $L(u) = \{\cos(u) - \cos(2u)\}/(\pi u^2)$, whose transform is 1 on $[-1, 1]$ and tapers linearly to zero at $|z| = 2$. Both converge quickly when f is smooth, and both misbehave in the manner described above.

2.2. The distance criterion

Doosti and Hall (2016) choose $\theta = (h, p)$ to minimise $\|\hat{f}(\cdot | \theta) - \check{f}\|$ in L_2 . Their Theorem 1 establishes that the minimiser inherits the convergence rate of $\check{f}$, and their Theorem 2 that it does so uniformly over classes of densities. The estimator is therefore rate-optimal in the sense of Stone (1980) while remaining a proper density.

2.3. The cross-validation criterion

Doosti *et al.* (2018) replace the comparator with

$$\text{CV}(h, p) = \int \hat{f}(x | h, p)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_{-i}(X_i | h, p), \quad (4)$$

where $\hat{f}_{-i}$ omits the term in X_i . As the authors note, p need not be re-standardised after the deletion, which saves computation. Minimising (4) over h and p jointly attains an integrated squared error of strictly smaller order than the $n^{-4/5}$ available to $\check{f}$.

3. One convex program

Write A for the $n \times n$ matrix with entries

$$A_{ij} = \int K_h(x - X_i) K_h(x - X_j) dx = (K_h * K_h)(X_i - X_j), \quad (5)$$

so that $\int \hat{f}(x | h, p)^2 dx = p^\top A p$ exactly. Both criteria then have the form

$$p^\top A p - \frac{2}{n} p^\top b + \text{const}, \quad (6)$$

and they differ only in b . For the criterion of Section 2.2, $b_i = \sum_j \int K_h(x - X_i) L_g(x - X_j) dx$, the cross terms against the comparator, and the constant is $\|\check{f}\|^2$. For the criterion of Section 2.3, $b_j = \sum_{i \neq j} K_h(X_i - X_j)$, obtained from the leave-one-out fit, and the constant is zero.

Minimising (6) subject to $p \geq 0$ and $\sum_i p_i = 1$ is a quadratic program over the probability simplex. It is convex: the Fourier transform of $K * K$ is $\phi_K^2 \geq 0$, and a function with a non-negative Fourier transform has a positive semidefinite Gram matrix. Equivalently, $p^\top A p = \int \hat{f}^2 \geq 0$ for every p . *This holds whatever kernel is used*, which is what allows the package to offer a choice of K without further argument.

The practical consequence is that the solution is unique and is found by a standard method: **tiltdens** uses the dual algorithm of Goldfarb and Idnani (1983) through **quadprog** (Turlach and Weingessel 2019), falling back to an accelerated projected-gradient scheme when the quadratic form is too ill-conditioned. For $m \leq 6$ variables an exact enumeration of the possible supports is faster still, and is used inside the search of Section 3.1.

3.1. Blocked weights and their boundaries

Section 4.1 of Doosti and Hall (2016) restricts the weights to m distinct values over contiguous blocks of the order statistics, and Algorithm A of Doosti *et al.* (2018) does the same with

approximately equal blocks. Writing $p = Gq$ for the $n \times m$ block indicator G , (6) becomes an m -dimensional program in q with $G^\top AG$ and $G^\top b$ in place of A and b .

The 2016 paper treats the block *boundaries* as part of the optimisation. Doing so naively is expensive, but the block sums of A can be read off a two-dimensional cumulative sum in constant time, and the resulting program has only m variables, so every admissible set of boundaries can be examined for $m \leq 3$. Above that the package refines the boundaries by coordinate descent from an equally spaced start.

3.2. Shape constraints

Since $\hat{f}$ is linear in p , a shape requirement evaluated on a grid $t_1 < \dots < t_M$ is a set of linear inequalities. With $\Phi_{kj} = K_h(t_k - X_j)$ the fitted curve is Φp , so monotonicity is $(\Phi_{k+1,\cdot} - \Phi_{k,\cdot})p \geq 0$, and unimodality with the mode at t_{k_0} is that condition up to k_0 and its reverse after. The program stays convex; only its feasible set gains rows. The mode location is unknown, so the program is solved for each candidate mode and the best solution retained, the device used by Hall and Huang (2002).

3.3. Multivariate data

Section 3.3 of Doosti and Hall (2016) replaces K with a product kernel $K_{\text{pr}}(x) = \prod_j K(x_j)$ and takes a scalar bandwidth. Every quantity the method needs then factorises,

$$\int K_{\text{pr},h}(x-u) L_{\text{pr},g}(x-v) dx = \prod_{j=1}^l \int K_h(x_j - u_j) L_g(x_j - v_j) dx_j, \quad (7)$$

so the l -variate matrices are elementwise products of the univariate ones and the program is unchanged. The weights remain an n -vector on the simplex.

3.4. Kernels

Nothing above requires K to be Gaussian. The Gram matrix (5) needs $K * K$, the cross terms need the Fourier transform ϕ_K , and evaluation needs K itself; given those three, the program is unchanged, and it is convex for every K by the argument following (6). The package provides the Gaussian; the k -fold convolutions of a Laplace density for $k = 1, \dots, 4$, whose transform is $(1+t^2)^{-k}$ and for which the smoothness condition of Section 3.1 of Doosti and Hall (2016) is stated; and the Epanechnikov, biweight, triweight and triangular kernels. For the Laplace family $K * K$ is the $2k$ -fold convolution and is available in closed form; for the compact kernels it is computed once by quadrature, piecewise between the points where it is not smooth, and interpolated.

A bandwidth is only meaningful relative to the kernel it scales. The package therefore rescales its default bandwidth by the canonical factor $\delta_K = \{R(K)/\mu_2(K)^2\}^{1/5}$ of Marron and Nolan (1988), where $R(K) = \int K^2$ is read off $K * K$ at zero and $\mu_2(K)$ is the second moment. With that rescaling the same numerical bandwidth smooths by the same amount whichever kernel is used; in the example of Section 4 the spread in integrated squared error across four kernels falls from about 2 to 1.1.

3.5. Numerical details

Two computations deserve care, because both are easy to get subtly wrong.

The cross terms. Every inner product between two kernel estimators reduces, by Parseval's identity, to an integral of the form $\int_0^c w(z) \cos(tz) dz$ with $t = (X_i - X_j)/h$, and there are n^2 of them. The integrand oscillates $t/2\pi$ times over the range, so a fixed quadrature rule fails once $|t|$ is large – that is, once the pairwise distance between observations is large relative to the bandwidth, which is the usual case in the tails. A fixed rule with a dozen or so nodes, for instance, is accurate to 10^{-6} at $|t| = 5$, to 4% at $|t| = 20$, and is meaningless beyond $|t| = 40$. The package scales the number of Gauss–Legendre panels with $\max |t|$, evaluates all n^2 integrals as one matrix product against the shared nodes, and is checked against the closed form $\int_0^1 \cos(tz) dz = \sin(t)/t$ at $|t| = 60$.

The comparator bandwidth. Least-squares cross-validation for an infinite-order kernel is multimodal in h , and its deepest minimum is frequently a spurious one at a very small bandwidth: beyond the frequency at which the empirical characteristic function decays into sampling noise, the criterion is fitting that noise. The package searches a frequency grid $q = 1/h$ and caps it at twice the frequency identified by the flat-top rule of Politis (2003), taking the minimum within that range. The cap is a judgement, documented and adjustable through an argument, and the criterion over the whole grid is returned so that the choice can be inspected.

4. Using the package

A bimodal sample serves throughout.

```
R> x <- c(rnorm(50, -1.5), rnorm(50, 1.5))
```

The recommended entry point is `tilt_density_cv`, which implements the criterion of Section 2.3.

```
R> fit <- tilt_density_cv(x)
R> fit
```

Tilted density estimate, cross-validated (Doosti, Hall & Mateu 2018)

```
Call:      tilt_density_cv(x = x)
Data:      x (100 obs.)
Bandwidth: 0.62462 (gaussian kernel)
Blocks:    3
Breaks:    at ranks 33, 67 (x = -1.32, 1.10), chosen by "equal"
CV:        -0.14876
Minimum:    7.8671e-05 (a proper density cannot go below zero)
```

Fitted objects inherit from class "density", so `plot`, `lines` and `points` behave as they do for `stats::density`. The point of the exercise is visible in two numbers.

```
R> c(sinc = min(sinc_density(x)$y), tilted = min(fit$y))

           sinc           tilted
-1.425296e-02  7.867101e-05
```

The comparator takes negative values; the tilted estimate cannot.

4.1. The distance criterion and block structure

`tilt_density` implements Section 2.2. The argument `m` sets the number of distinct weights and `breaks` how their boundaries are placed: "equal" is Algorithm A, "modal" places boundaries at the troughs of a pilot estimate, and "optimal" searches them as in Section 3.1.

```
R> sapply(c("equal", "modal", "optimal"), function(b)
+   tilt_density(x, m = 3, breaks = b)$distance2)
```

	equal	modal	optimal
	0.001974659	0.002116792	0.001629076

4.2. Kernels and canonical bandwidths

The kernels of Section 3.4 are selected by name.

```
R> tilt_kernels()
```

```
[1] "gaussian"      "laplace"      "laplace2"    "laplace3"
[5] "laplace4"      "epanechnikov" "biweight"    "triweight"
[9] "triangular"
```

```
R> round(sapply(c("gaussian", "epanechnikov", "biweight"),
+   bw_canonical_factor), 4)
```

gaussian	epanechnikov	biweight
0.7764	1.7188	2.0362

4.3. Shape constraints and multivariate data

```
R> tilt_density(x, m = Inf, constraint = "unimodal")
```

```
R> X <- cbind(x, c(rnorm(50, -1), rnorm(50, 1)))
R> plot(tilt_density(X))
```

5. Illustrations

5.1. Simulation

A brief simulation, in the design of both papers, shows where the perturbed estimators help and where they do not. Four of the eight normal-mixture densities of Marron and Wand (1992) are

Density	Conv.	Sinc	Trap.	T_n	T_3	CV
Gaussian	5.81	6.14	6.89	5.73	5.77	6.76
Kurtotic unimodal	104.92	75.82	60.77	74.09	59.03	68.63
Separated bimodal	43.67	15.22	16.66	13.48	14.31	27.20
Skewed bimodal	11.38	16.61	13.09	15.63	13.22	12.22

Table 1: Mean integrated squared error, multiplied by 1000, for $n = 100$ over 40 replicates. The smallest value in each row is in bold. Conv.: conventional Gaussian kernel estimator; Trap.: trapezoidal estimator; T_n and T_3 : tilted estimators with n and 3 distinct weights; CV: the cross-validation estimator.

used, chosen to span the range from simple to structured: the Gaussian, the kurtotic unimodal (a spike on a broad base), the separated bimodal, and the skewed bimodal. Samples are of size $n = 100$. The estimators compared are the conventional Gaussian kernel estimator with a robust normal-reference bandwidth; the sinc and trapezoidal estimators with a cross-validated bandwidth; the tilted estimator of Section 2.2 with one weight per observation (T_n) and with three distinct weights and optimised boundaries (T_3); and the cross-validation estimator of Section 2.3 with up to three distinct weights. The replication script reproduces Table 1 in about ten minutes.

Table 1 shows the pattern both papers report, and we do not wish to oversell it. Where the density is simple and smooth, the conventional estimator is hard to beat, and the perturbed estimators roughly match it. Where the density has sharp structure – a spike at the mode, well-separated modes – the perturbed estimators are clearly better, because they can follow the comparator into that structure without following it below zero. The tilted estimators never do worse than the comparator they are built from, which is the content of Theorem 1 of Doosti and Hall (2016) expressed numerically.

5.2. Old Faithful

Both papers analyse eruption durations of the Old Faithful geyser, using the 107 observations discussed by Silverman (1986). We use the 272 durations in R’s `faithful` data set, which come from a later collection period; the substantive question – is the density bimodal? – is the same.

```
R> x <- faithful$eruptions
R> fit_cv <- tilt_density_cv(x)
R> fit_sinc <- sinc_density(x, bw = fit_cv$bw, from = 1, to = 6)
R> c(sinc = min(fit_sinc$y), tilted = min(fit_cv$y))
```

```
      sinc      tilted
-0.053137201  0.000220287
```

Figure 1 shows the three estimates. The sinc estimator’s minimum of -0.053 is not a small defect: it is comparable in magnitude to the density in the trough between the modes. The tilted estimate has the same bandwidth and the same two modes, and its minimum is 2.2e-04.

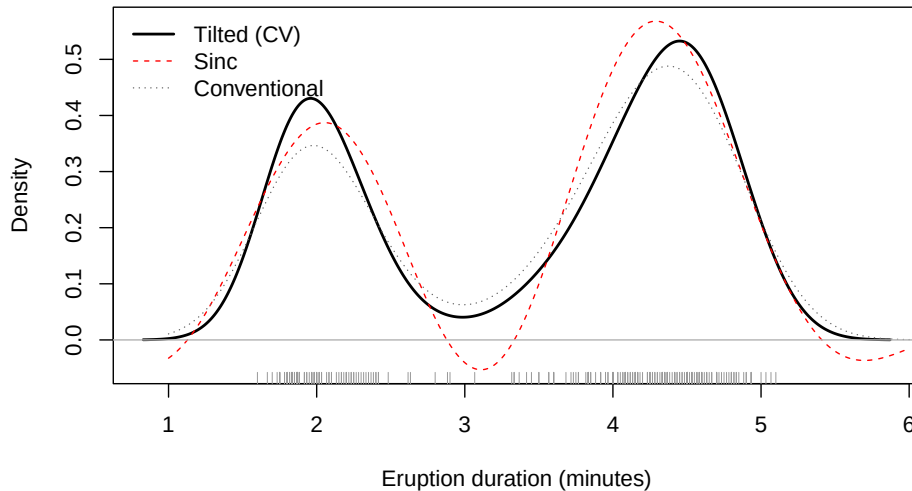


Figure 1: Old Faithful eruption durations. The sinc estimator dips well below zero between and beyond the modes; the tilted estimate tracks it where it is sensible and cannot follow it below zero. The conventional estimator, with a normal-reference bandwidth, is the most reluctant of the three to separate the modes.

5.3. Galaxy velocities

The 82 galaxy velocities of [Roeder \(1990\)](#), available as `galaxies` in `MASS`, are the standard test case for multimodality: the astrophysical question is whether the galaxies cluster, which would show as several modes. Shape constraints give a direct way to ask it.

```
R> g <- MASS::galaxies / 1000
R> free <- tilt_density(g, m = Inf)
R> unimodal <- tilt_density(g, m = Inf, constraint = "unimodal")
R> c(free = free$distance2, unimodal = unimodal$distance2)
```

```
      free      unimodal
0.001068358 0.005765238
```

Forcing the estimate to be unimodal moves it 5.4 times further from the infinite-order comparator. That ratio is an informal measure of how strongly the data resist a single mode, in the spirit of the bandwidth test of [Silverman \(1981\)](#) but reading the answer off a distance rather than a critical bandwidth.

6. Comparison with existing packages

Table 2 places `tiltdens` among the R packages a user might reach for. The comparison is not that `tiltdens` is better than `ks` at what `ks` does; it is that `tiltdens` does something none of the others does. The combination in the last column – an estimate that converges at a higher-order rate *and* is guaranteed to be a density – is not available elsewhere.

Two of the alternatives deserve a word. `logcondens` ([Dümbgen and Rufibach 2011](#)) also delivers a non-negative, shape-restricted estimate, but by maximising a likelihood over log-concave

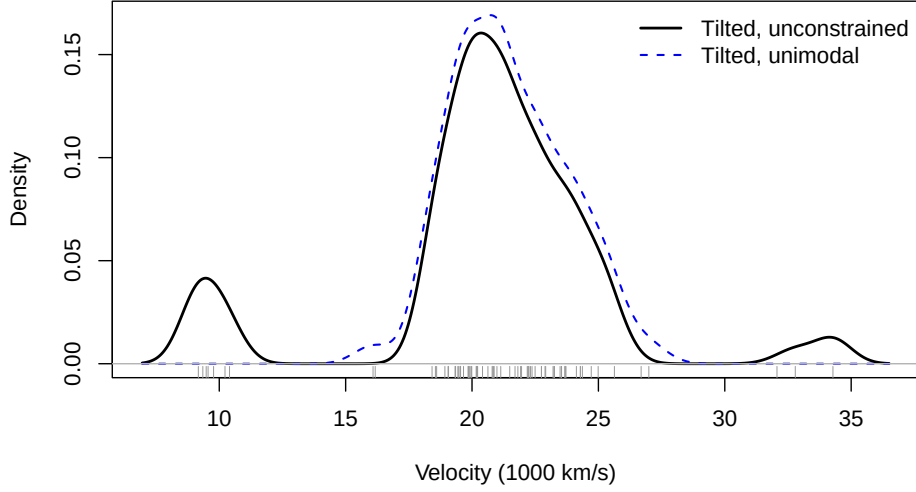


Figure 2: Galaxy velocities. The unconstrained tilted estimate shows the main cluster and the outlying groups; the unimodal fit is what the data would have to look like if there were one cluster, and it fits noticeably worse.

Package	Higher-order rate	Non-negative	Shape constraints	Multivariate
ks	yes (higher-order kernels)	no	no	yes
KernSmooth	yes (higher-order kernels)	no	no	bivariate
kdensity	no	yes	no	no
logcondens	no	yes	log-concavity	no
sharpData	regression only	—	—	—
tiltdens	yes (by perturbation)	yes	yes	yes

Table 2: Capabilities of R packages for density estimation. **sharpData** implements data sharpening for local linear regression (Choi *et al.* 2000), not density estimation.

densities; it is the right tool when log-concavity is a credible assumption, and the wrong one when the density is multimodal, as in Section 5.3. **ks** (Duong 2007) is the most complete general-purpose package and offers higher-order kernels, but a user who wants their rate must accept an estimate that goes negative; that is the trade-off **tiltdens** removes.

7. Summary and discussion

tiltdens implements the tilted and data-sharpened density estimators of Doosti and Hall (2016) and the cross-validation criterion of Doosti *et al.* (2018), which previously had no implementation in any language. The estimators inherit the convergence rate of an infinite-order kernel while remaining proper densities. The observation that makes them practical is that both selection criteria reduce to a single convex quadratic program over the probability simplex, differing only in one linear term: that gives a unique solution, replaces a search over a high-dimensional simplex with a standard solver, and extends without modification to blocked weights with optimised boundaries, to shape constraints, and to multivariate data. A choice

of kernel is offered, with bandwidths rescaled so that the choice affects the shape of the estimate rather than its degree of smoothing.

Several limitations should be stated plainly. Data sharpening is implemented for univariate data only, and it is slow: the objective is not convex in the shifts, so a stochastic search is used, and the papers' own finding is that tilting is usually at least as accurate and far cheaper. Block structure rests on the order statistics and has no natural multivariate counterpart, so multivariate fits use one weight per observation. Shape constraints are imposed on a grid, and a violation between grid points is possible in principle; in practice it is of relative order 10^{-5} and the grid can be refined. The cross-validation criterion for the comparator bandwidth is multimodal, and its global minimum is often a spurious one at a very small bandwidth, so the search is capped at a multiple of the frequency beyond which the empirical characteristic function is indistinguishable from noise; that cap is a judgement, documented and adjustable. Two extensions are natural. Doosti *et al.* (2018) report that tilted estimators for spatially correlated data were under investigation, and the machinery here – a linear estimator whose weights lie on the simplex – carries over to that setting once the criterion is adapted. And the shape constraints of Section 3.2 are only those that are linear in the weights on a grid; constraints on moments, on tail behaviour, or on the number of modes are all expressible in the same form, and adding them is a matter of assembling the inequality matrix.

Acknowledgements

Peter Hall died in January 2016, before the second of the two papers was complete; the authors of that paper dedicated it to him, and this package is dedicated to him likewise. The methods it implements are his as much as anyone's.

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